

FloodGuard Miami — Executive Summary

FloodGuard Miami is an innovative, no-code flood prediction and alert platform designed to help residents, planners, and local agencies prepare for urban flooding in Miami, Florida. Leveraging open-source data and AI-driven insights, FloodGuard Miami transforms static flood maps into real-time, predictive visualizations that empower decision-makers and protect communities.

Problem

Miami faces chronic flooding from heavy rainfall, king tides, and sea level rise. Existing FEMA and NOAA flood maps are static and often fail to reflect on-the-ground realities. Residents and planners lack accessible real-time tools to anticipate flooding and act proactively.

Solution

FloodGuard Miami integrates open data from NOAA, USGS, and local IoT sensors to deliver predictive flood mapping. Through a simple dashboard interface, users can view flood risks for the next 24–48 hours, receive alerts, and contribute local observations. The platform requires no coding and uses community-driven reporting to enhance accuracy and local trust.

Market and Go-to-Market Strategy

Our target market includes local governments, resilience offices, and coastal communities across South Florida. FloodGuard Miami begins with a pilot deployment in high-risk areas like Little River and Brickell, in collaboration with Florida International University (FIU) and the Miami-Dade Office of Resilience. The platform follows a freemium model — offering free tools for residents and premium analytics for municipalities and insurers.

Value Proposition

By delivering hyperlocal, predictive flood data, FloodGuard Miami helps communities reduce property damage, improve urban resilience planning, and foster citizen engagement. Our approach bridges the gap between scientific research and real-world adaptation needs in one of America's most flood-prone regions.

Founder

Mostafa Sameer is a Graduate Research Assistant at the Department of Civil and Environmental Engineering, Florida International University (FIU). His research focuses on flood risk mitigation and resilient infrastructure. He founded FloodGuard Miami to turn academic research into actionable community tools for Miami's flood resilience.

References

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